A Cost-Effective LiDAR–Inertial Device for 2D Cave Surveying in Ferromagnetic Environments

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***Abstract*—Cave surveys operate without GNSS and accumulate distance, inclination, and azimuth between stations. Commercial smart surveyors support digital mapping, but magnetometer-based heading degrades under local magnetic disturbance, common in iron-rich rock and under weak regional fields such as the South Atlantic Anomaly. This paper presents a cost-effective open cave survey instrument with an embedded attitude and heading reference system (AHRS), a millimeter-class laser rangefinder, onboard survey logging, and Bluetooth Low Energy connectivity to digital sketching apps, i.e., mobile software that receives each shot and draws the cave map on a phone or tablet. Experiments include attitude evaluation against optical motion capture, a magnetic-stress test against a mechanical compass, and a short field traverse versus a commercial smart surveyor in the same sketching workflow. Results show sub-degree mean absolute error in inclination and roll, azimuth errors of a few degrees after extrinsic bias removal, and heading that remains usable under magnet stress that deflects a mechanical compass. In the field, the proposed instrument recovers the same sketch topology as the commercial reference on a paired traverse. Overall, the results indicate an affordable open tool that can be compared with established smart-surveyor solutions already used for 2D cave surveying, but supporting ferromagnetic environments.**

***Index Terms*—Cave Surveying, Speleology, Inertial Measurement Unit, Magnetic Anomalies, Robotics Hardware.**

I. Introduction

Speleological mapping and topographical documentation support geological studies [1], biodiversity monitoring [2], and impact assessment for mining and infrastructure [3]. Underground, GNSS is unavailable, so surveys use dead-reckoning traverses: at each station the operator measures a *shot* (also called a *leg*) to the next station as distance, inclination (elevation angle of the laser), and azimuth (heading relative to magnetic north) [4]. Traditional tools were a tape, magnetic compass, and clinometer; sketching was done on waterproof paper.

Over the past two decades, the community moved to handheld electronic units often called smart surveyors (in Portuguese, *trenas* eletrônicas) [5], [6]. A smart surveyor integrates a laser rangefinder with an attitude and heading reference system (AHRS) based on accelerometers, gyroscopes, and magnetometers, then transmits each shot wirelessly to a phone or tablet for on-site digital sketching and topological checks, for example with TopoDroid on Android [7].



Fig. 1. Proposed handheld cave survey instrument (smart surveyor) in an underground passage.

Commercial units remain limited by cost, proprietary architectures, and magnetic anomalies. Heading uses the local geomagnetic field, which iron-rich rock can distort [8], [9]. Under a weak regional field such as the South Atlantic Anomaly [10], [11], local anomalies more easily dominate the compass reference and degrade conventional compasses and AHRS algorithms [12], [13].

This paper presents a cost-effective, open-architecture smart surveyor[1] for magnetically disturbed and ferromagnetic environments, including iron-rich caves. The contributions comprise a hardware and firmware stack (embedded AHRS, laser ranging, onboard survey logging, and Bluetooth Low Energy connectivity to digital sketching software) with a rationale for the chosen microcontroller (MCU), sensor hub, laser, and enclosure; and an experimental validation with optical motion-capture attitude tests, a magnetic-stress comparison against a mechanical compass, and a short field survey against a commercial smart surveyor in the same sketching workflow.

The remainder of this paper is organized as follows. Section II reviews related cave surveying instruments and software. Section III presents design decisions and a bill of materials, the mechanical and electrical architecture, and

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[1]Project page: https://mira-robotica.github.io/mm1-black/.

the firmware, including calibration and filtering. Section IV presents the experimental validation. Section V concludes.

II. Related Work

Cave surveying traditionally used tape, compass, and clinometer (e.g., Suunto Tandem), with sketches on waterproof paper. Heeb catalyzed *paperless caving* [5], [14]: one handheld unit measures a shot and transmits it wirelessly for on-site sketching [15]. Early electronic platforms such as the Shetland Attack Pony (SAP), DUSI, and CaveSniper added electronic compass/clinometer sensing, initially without or with limited laser ranging [14], [16], [15].

The DistoX board fitted inside a Leica DISTO A3 combined a laser rangefinder with accelerometers, magneto-inductive sensors, and Bluetooth 2.0 Serial Port Profile (SPP) [14], [17]. DistoX2 (Leica X310) improved sensing and firmware upgrades, with compass accuracy near 0.5° RMS after a 14×4 calibration (fourteen orientation groups with four rolls each) in PocketTopo or TopoDroid [18], [19], [20]. PocketTopo is a Windows application for receiving shots and drawing cave surveys; TopoDroid is an Android application for the same digital sketching workflow on a phone or tablet [7], [21]. Production ended after Leica discontinued the host DISTO models [19]. Field tests with total stations confirmed usability in caves while exposing residual angular discrepancies [22].

SAP5 and SAP6 are commercial laser+compass surveyors with Bluetooth transfer to Android. SAP6 quotes approximately $\pm0.5^\circ$ compass, $\pm0.3^\circ$ clinometer, and ±1 mm laser when calibrated, and defines CaveBLE/SAP6: a Bluetooth Low Energy (BLE) Generic Attribute Profile (GATT) that encodes survey shots so apps can subscribe to leg notifications [19], [23]. BRIC4 and BRIC5 are commercial smart surveyors with redundant sensors, on-device calibration, SD logging, and BLE; BRIC5 reports azimuth accuracy better than 0.5° and distance accuracy of a few millimeters [24], [25], [19]. Field surveys with BRIC5 and TOPGRU, a Brazilian club workflow for drafting cave maps from electronic shots, report 1.1–1.5% traverse closures under moderate magnetic interference [26]. Related options include DiscoX, an open-hardware SAP6-compatible surveyor [27], Cavway X1, a proprietary BLE unit [28], and mobile solid-state LiDAR as a volumetric alternative to station-to-station surveying [19].

Table I consolidates published attributes. Across this landscape, azimuth remains magnetometer-dependent and sensitive to local anomalies [12], [29]. The proposed instrument targets a cost-effective open architecture, BLE interoperability with digital sketching software, and laboratory evidence of heading under magnetic stress. The paired field check addresses workflow and sketch consistency rather than metrological equivalence (Section IV).

III. System Architecture

The proposed instrument combines a compact mechanical enclosure, a sensing and processing electronics stack, and firmware for standalone CSV logging and wireless digital surveying.

A. Design Decisions and Bill of Materials

Component selection follows three constraints for research and club-level use: cost-effective unit price, repairability with commodity parts, and interoperability with digital sketching software. These constraints favor modules that operators can source, replace, and reprogram without vendor lock-in.

The host MCU is an ESP32 (dual-core, up to 240 MHz, 520 KB SRAM) with on-chip BLE, which avoids a separate radio. One core runs LVGL, a lightweight graphics library for embedded GUIs, and touch events; the other handles sensor polling, CSV writes, and BLE pacing. An integrated ESP32 board with a 4.0 inch ST7796 TFT, touch, and microSD reduces assembly and provides an editable survey table underground.

Attitude estimation uses the BNO086 sensor hub rather than a raw inertial measurement unit (IMU) with a host-side Madgwick or complementary filter. Madgwick-type filters fuse accelerometer, gyroscope, and magnetometer data on the host MCU; the BNO086 instead runs CEVA SH-2, an on-chip sensor-fusion engine, exposes a rotation-vector quaternion at tens of hertz, and reports a fusion-quality indicator under magnetic disturbance [30]. Distance sensing uses an LDL UART laser module (0.03–40 m, millimeter-class accuracy, 3.3 V TTL), which covers common cave legs. Mechanics use a two-part printed shell, a 2000 mAh LiPo pack for several hours of intermittent surveying, and a $1/4''$-20 insert for tripod mounting. Communication adopts the CaveBLE/SAP6 GATT profile so that sketching apps that already support BRIC and SAP devices can receive shots without a proprietary stack [23].

Table II summarizes an approximate retail bill of materials for one prototype (USD, early 2026). Electronics and enclosure total about $140–160, below many commercial smart surveyors for the same digital sketching workflow.

B. Electrical and Mechanical Design

Figure 2 shows the assembled prototype (top and front), and Fig. 3 shows the exploded layout of the main subsystems. The enclosure is a two-part 3D-printed shell for handheld underground use, with a wrist strap and a standard $1/4''$-20 insert for monopod or tripod mounting. The laser aperture and measure button sit together on the front face to support a comfortable aiming posture in confined passages.

Figure 4 summarizes the electrical interconnection. The LiPo pack feeds the ESP32 board regulation stage; USB Type-C supports charging and firmware update. The BNO086 communicates over I^2C, the laser module uses UART, and the display and microSD share SPI buses. Measure and power buttons connect to GPIO, and the ESP32 exposes BLE for digital sketching clients.

C. Software and Firmware Design

The firmware coordinates the touch interface, attitude estimation, laser acquisition, local comma-separated value (CSV) storage of survey legs, and BLE transfer to sketching software. In the experiments of this paper the client is TopoDroid [7], [21]. Figure 5 summarizes the layered stack, the shot-delivery path, and the AHRS data path. The survey loop is aim, acquire

TABLE I
COMPARISON OF ELECTRONIC CAVE SURVEYING TOOLS (MANUFACTURER OR PUBLISHED FIGURES)

Device	Sensors	Link / apps	Distance	Angular (typ.)	Calibration	Arch.
Proposed	LDL 40 m laser; BNO086 9-DoF	BLE SAP6; sketching apps	± 1–2 mm	few° az. (lab); $< 1^\circ$ inc.	hub + az. offset	Open
DiscoX [27]	ToF laser; 9-DoF IMU	BLE Coded-PHY (SAP6)	± 1 mm	$\pm 0.5^\circ$	onboard	Open
Cavway X1 [28]	Laser ToF; 9-DoF IMU	BLE; TopoDroid	± 1 mm	$\sim \pm 0.5^\circ$	onboard	Prop.
BRIC5 [24], [25]	LDK-M2; 2×SCA3300, 2×RM3100	BLE 4.2; TopoDroid	1/3 mm	az. $< 0.5^\circ$; inc. $< 0.2^\circ$	14×4 onboard	Prop.
BRIC4 [16]	Laser; red. accel. + mag.	BLE; CSV/SD; TopoDroid	~ 1 mm	az. $\approx \pm 0.20^\circ$	14×4 onboard	Prop.
SAP6 [19]	Laser + 9-DoF	BLE SAP6; TopoDroid	± 1 mm	$\pm 0.5^\circ$ az.; $\pm 0.3^\circ$ inc.	onboard	Prop.
SAP5 [16]	Laser + compass/clino	BT; TopoDroid	mm-class	below DistoX2/BRIC	onboard (2020)	Prop.
DistoX2† [18]	Leica X310; PNI mag. + dual accel.	BT 2.0 SPP; TopoDroid	$\sim \pm 1$ mm	$\sim 0.5^\circ$ RMS	14×4 external	Prop.
DistoX† [14], [17]	Leica A3; PNI mag. + SCA3000	BT 2.0 SPP; PocketTopo	$\sim \pm 1$ mm	mag. sens. 0.03°–0.1°	14×4 external	Prop.
CaveSniper [15], [16]	Laser + electronic compass/clino	BT; vendor PC app	mm-class	survey-grade (vendor)	device/PC	Prop.

†Out of production or superseded. Early platforms (DUSI) and mobile LiDAR meshes are omitted for space; see [14], [19].

TABLE II
APPROXIMATE BILL OF MATERIALS FOR ONE PROTOTYPE (RETAIL ESTIMATES)

Item	Qty	Unit (USD)	Ext. (USD)
ESP32 + 4.0″ TFT (CYD/ST7796)	1	20–30	25
BNO086 9-DoF AHRS breakout	1	30–35	35
LDL 40 m laser (UART)	1	45–60	50
LiPo 3.7 V, 2000 mAh	1	6–10	8
microSD card (survey CSV)	1	4–8	5
Buttons, USB-C, wiring	1 lot	–	8
3D-printed enclosure	1	–	8
1/4″-20 insert, strap, fasteners	1 lot	–	5
Approximate total			**∼140–150**

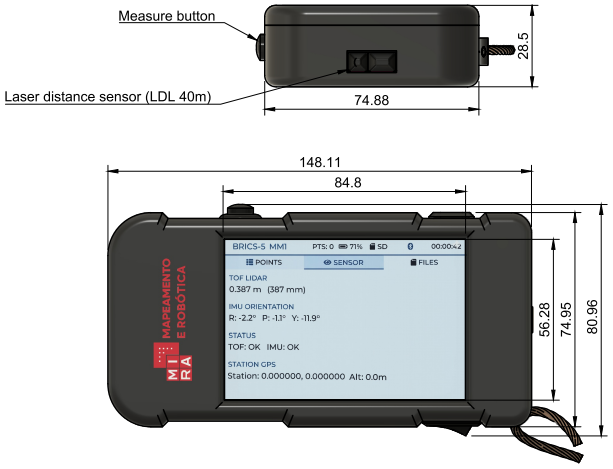


Fig. 2. Mechanical drawing of the proposed survey tool (top and front views), showing overall dimensions, 4.0 inch TFT interface, measure button, and LDL laser aperture. All linear dimensions are in millimeters (mm).

a synchronized distance/attitude shot, store the leg locally, and optionally deliver it under ACK pacing.

LVGL implements the user interface with four primary tabs: POINTS (editable survey table), SENSOR (live distance and orientation), FILES (CSV session management on the microSD card), and SETUP (brightness, calibration, Bluetooth diagnostics, and device information). Capture follows a two-press workflow used by commercial smart surveyors [16]: the first press enables only the aiming laser, and the second press

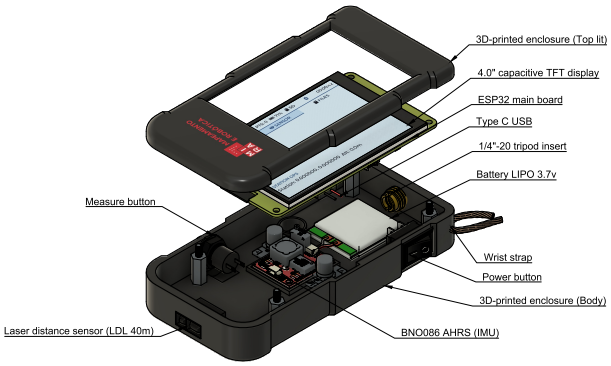


Fig. 3. Exploded view: 3D-printed enclosure, TFT/ESP32 stack, BNO086, LDL laser, LiPo pack, USB-C, buttons, wrist strap, and 1/4″-20 insert.

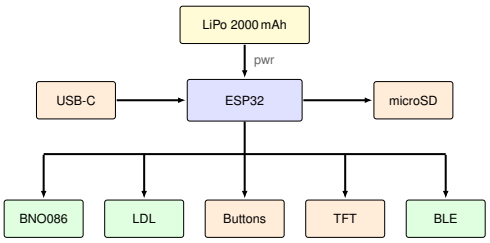
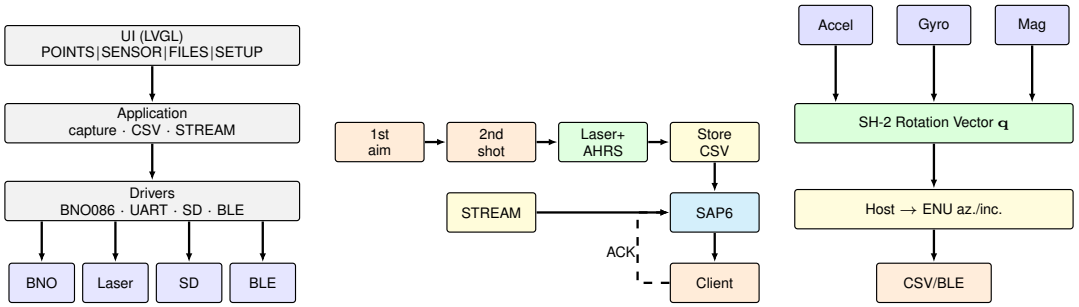


Fig. 4. Simplified electrical interconnection of the proposed survey tool.

triggers a synchronized laser reading and freezes the current AHRS attitude. The firmware appends the resulting leg to a survey CSV file and, when a client connects, notifies it over BLE.

a) Attitude filtering.: The BNO086 SH-2 *Rotation Vector* report supplies orientation, typically at 50 Hz, with on-chip fusion of accelerometer, gyroscope, and magnetometer data and dynamic magnetic disturbance handling [30]. The ESP32 host does not run a second Madgwick-type filter; it rotates a calibrated laser-axis body vector through the quaternion into an East–North–Up (ENU) frame and extracts azimuth and inclination (Fig. 5c). An accelerometer magnitude check ($\approx 9.8\,\mathrm{m/s^2}$) flags leveling errors before storage.

b) Calibration procedures.: SETUP exposes a lighter field calibration than the DistoX/BRIC 14×4 roll-group procedure. Power-on and hub calibration must be done away from massive metal: starting the AHRS next to a large metallic



(a) Firmware layers (b) Shot / BLE delivery (c) AHRS path

Fig. 5. Firmware organization: (a) software layers; (b) two-press shot acquisition and ACK-paced SAP6 delivery; (c) on-chip fusion and host angle extraction.

object distorts azimuth, whereas removing the source lets the filter reconverge. For cave work, complete this step outdoors or at the entrance until the BNO086 reports stable "good" or "excellent" fusion quality. A persistent azimuth/declination offset $\Delta\psi$ in non-volatile memory aligns heading with magnetic or grid north, and a fixed body-frame laser direction vector corrects mounting offset for a given enclosure.

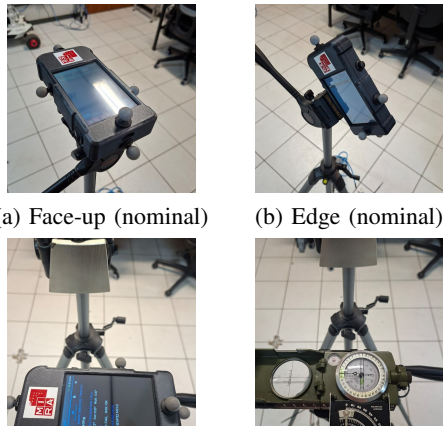
c) Data integrity and BLE link.: Wireless transfer uses the CaveBLE/SAP6 GATT profile [23], [27]: the instrument notifies each shot as a short leg packet; the phone acknowledges before the next packet (ACK-paced delivery with retransmission). STREAM mode can replay stored CSV legs after disconnected surveying. Recent points stay in an editable RAM window for the on-screen table, while older measurements spill to the SD card.

IV. Experimental Validation and Results

This section presents experiments that evaluate attitude precision and the digital survey workflow. The laboratory study measures attitude error against optical motion capture under nominal conditions and under magnetic stress, with a military compass as a classical reference; it does not include a commercial BRIC unit (Section IV-A). The field study compares the proposed instrument with a commercial smart surveyor (BRIC-5) on a short outdoor traverse streamed to TopoDroid (Section IV-B). Details of each protocol follow.

A. Attitude Characterization Against Optical Motion Capture

Optical motion capture provides a magnetically independent orientation ground truth. An analog military compass aligned the capture frame to magnetic north before the session, and a laboratory azimuth offset of about 30° entered the conversion from capture quaternions to the survey angles reported by the instrument. The survey angles are azimuth (heading about vertical), inclination (elevation of the laser axis), and roll about the laser axis; they are derived from the attitude quaternion and are the quantities stored in each shot, rather than a generic body-fixed Euler triple. A tripod with a three-degree-of-freedom support and reflective markers held the device rigidly (Fig. 6). The protocol covered faces and edges of the enclosure; for each vertically oriented face or edge, the setup recorded eight azimuth stops spaced about 45° from the laboratory reference. The operator ran the hub calibration



(a) Face-up (nominal) (b) Edge (nominal)

(c) Instrument + magnet (d) Compass + magnet

Fig. 6. Laboratory setups: (a)–(b) optical motion-capture mounts; (c)–(d) magnetic-stress test with a strong permanent magnet on the instrument and on a military compass/clinometer.

outdoors, away from metal (figure-eight, roughly 30 s), before indoor logging, and repeated a short laboratory recalibration after battery recharge or power cycles.

ROS 2 bags stored continuous motion-capture streams, while the instrument SD log stored discrete shots time-aligned to steady orientation windows. The analysis keeps only plateau interiors. Let a_{inst} be the instrument reading and a_{mocap} the motion-capture reference for the same survey axis (azimuth, inclination, or roll), both in degrees. The signed circular error is

$$e = \big((a_{\mathrm{inst}} - a_{\mathrm{mocap}} + 180^\circ) \bmod 360^\circ\big) - 180^\circ, \quad (1)$$

where $e \in (-180^\circ, 180^\circ]$ wraps the angular difference onto that interval. The median of e at each stop is retained. Removing a global median bias per axis suppresses a constant extrinsic offset between the rigid body and the instrument frame. Near-vertical mounts (positions 13 and 17) make azimuth and roll singular when extracted from the attitude chart, so the analysis excludes those axes there and keeps inclination.

Table III summarizes bias-corrected errors over $N{=}121$ azimuth/roll stops and $N{=}123$ inclination stops in the multi-mount nominal campaign. Inclination and roll stay near 1° MAE (0.65° and 0.83°), while azimuth dominates (MAE 2.90°, RMS 3.76°, p_{95} $|e|{=}6.89^\circ$), as expected for

TABLE III
ATTITUDE ERROR VS. OPTICAL MOTION CAPTURE UNDER NOMINAL CONDITIONS AFTER REMOVING A GLOBAL PER-AXIS MEDIAN BIAS. CIRCULAR DIFFERENCES; AZIMUTH/ROLL EXCLUDED AT NEAR-VERTICAL POSES (POSITIONS 13 AND 17). EACH SAMPLE IS THE MEDIAN ERROR OF ONE STEADY STOP.

Axis	**N**	**MAE** (°)	**RMS** (°)	$\mathbf{p}_{95}$ $\lvert e\rvert$ (°)	**Max** $\lvert e\rvert$ (°)
Azimuth	121	2.90	3.76	6.89	16.08
Inclination	123	0.65	0.77	1.36	1.91
Roll	121	0.83	1.07	2.27	2.87

magnetometer-aided heading even without intentional magnetic stress.

Figure 8 decomposes azimuth $|e|$ by mount. Errors are uneven: mounts 16, 4, and 8 reach about 4°–6° MAE, while mounts 2, 3, and 6 stay near 1.6°–1.8°. Inclination remains low across mounts, which points to heading residuals from magnetometer and soft-iron interaction with the metal tripod and lab field rather than accelerometer leveling.

a) Magnetic stress test.: A fair comparison with the magnet session uses a geometrically matched baseline: the magnet run is an eight-stop yaw sweep with the device nearly level, so the reference subset is the nominal stops on level mounts only (positions 1–7 and 15, N=64). For each session, *Raw MAE* is the mean absolute difference versus motion capture, and *Residual MAE* removes that session's median offset.

Table IV and Fig. 7 summarize the result. Raw errors under the magnet stay close to the level nominal baseline (azimuth 2.43° vs. 2.55°). The residual azimuth error drops from 2.34° to 0.15°, i.e., the magnet mainly adds a nearly constant offset. Qualitatively, the same magnet deflects a military compass/clinometer (Fig. 6d) while the AHRS remains usable on the UI (Fig. 6c), provided the hub was calibrated away from metal.

TABLE IV
LEVEL-MOUNT BASELINE VS. MAGNETIC STRESS (PLATEAU MEDIANS VS. OPTICAL MOTION CAPTURE). NOMINAL (LEVEL): POSITIONS 1–7 AND 15. RESIDUAL MAE REMOVES THE PER-SESSION MEDIAN OFFSET.

Condition	**Axis**	**N**	**Raw MAE** (°)	**Resid. MAE** (°)
Nominal (level)	Azimuth	64	2.55	2.34
Nominal (level)	Inclination	64	0.68	0.33
Nominal (level)	Roll	64	0.70	0.53
Magnet	Azimuth	8	2.43	0.15
Magnet	Inclination	8	0.63	0.17
Magnet	Roll	8	0.73	0.28

B. Field Comparison in a Digital Sketching Workflow

Figure 9(a) shows the proposed instrument during underground capture. A commercial smart surveyor (BRIC-5) and the proposed tool then surveyed the same outdoor closed traverse, each streaming legs over BLE (SAP6) to TopoDroid. Figures 9(b)–(c) show the resulting plan sketches, and Table V lists the sketch statistics reported by the app. Both pipelines

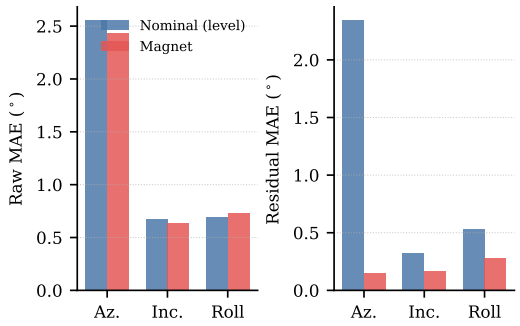


Fig. 7. Level nominal vs. magnet stress: raw MAE (left) and residual MAE after removing the per-session median offset (right).

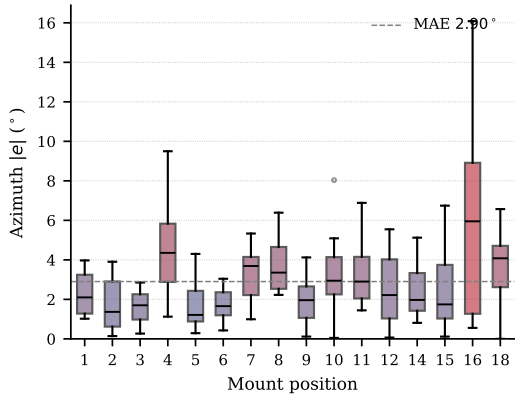
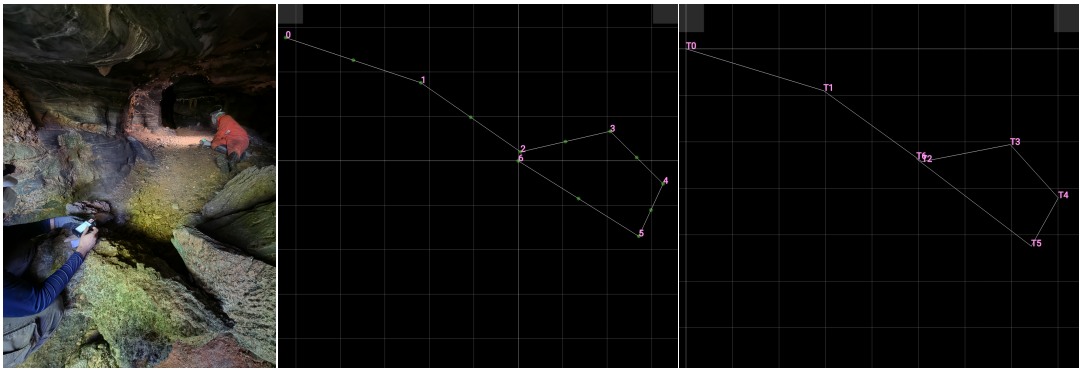


Fig. 8. Azimuth absolute error by mount position (bias-corrected). Dashed line: overall MAE. Positions 13 and 17 omitted (singularity).

recover the same topology (approach legs into a four-station loop that returns near the junction station). Each survey used six shots and seven stations, with total plan length 14.2 m (proposed) and 13.5 m (commercial). The sketching software reports mean shot-position errors of 2.3 cm and 1.5 cm, and mean azimuth errors of 0.29° ± 0.13° and 0.09° ± 0.05°, respectively. These values are internal sketch-consistency indicators, not an external ground truth, and the commercial unit remains tighter on that metric. The comparison supports BLE workflow compatibility and topological agreement on a short paired traverse, without catalog-grade equivalence to BRIC-5. TopoDroid acknowledged shots without protocol failures.

V. CONCLUSION

This paper presented a cost-effective open cave survey instrument for ferromagnetic and magnetically disturbed environments, combining an ESP32 host, embedded AHRS, LDL laser, onboard survey logging, and BLE connectivity to digital sketching software at about $140–160 per prototype. Optical motion-capture tests showed sub-degree inclination/roll errors and azimuth errors of a few degrees after extrinsic bias removal, which is coarser than published commercial azimuth claims near 0.5°. Under a magnet that deflects a military compass, level-mount raw errors stayed close to the nominal baseline while azimuth residuals remained nearly constant



(a) Cave capture (b) Proposed (TopoDroid) (c) BRIC-5 (TopoDroid)

Fig. 9. Field workflow: (a) underground capture with the proposed smart surveyor; (b)–(c) TopoDroid plan sketches of the same outdoor closed traverse streamed over SAP6 BLE from the proposed instrument and from BRIC-5.

TABLE V
TOPODROID SKETCH STATISTICS FOR THE PAIRED OUTDOOR TRAVERSE

Quantity	**Proposed**	**Commercial**
Shots / stations	6 / 7	6 / 7
Plan length (m)	14.2	13.5
Mean shot position error (cm)	2.3	1.5
Mean azimuth error (°)	0.29 ± 0.13	0.09 ± 0.05

across stops. In the field, the instrument recovered the same sketch topology as a commercial smart surveyor on a short outdoor traverse, with looser internal sketch consistency and no external ground truth. The results therefore validated a usable open workflow and laboratory attitude behavior, not metrological parity with catalog commercial surveyors. Future work includes field campaigns in iron-rich caves under different magnetic conditions; multi-axis or dual-laser ranging for left/right wall shots; compact low-cost 2D LiDAR for cross sections; optional gas sensing; and GNSS at entrances with propagation of an external georeference into the subterranean network.

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